LAB 7

Introduction to Git, GitHub, and GitHub Actions

Part A – Git Basics

Task A1: Initialize Repository

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~
$ cd desktop

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop
$ mkdir devops-lab

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop
$ cd devops-lab

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab
$ git init
Initialized empty Git repository in C:/Users/SIDDESHWARAPRASAD KR/Desktop/devops-lab/.git/
```

Task A2: Create a Sample File

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ echo "Hello DevOps Lab" > index.txt
```

Task A3: Stage and Commit

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git add .
warning: in the working copy of 'index.txt', LF will be replaced by CRLF the next time Git touches it

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git commit -m "Initial Commit"
Author identity unknown

*** Please tell me who you are.

Run

  git config --global user.email "you@example.com"
  git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO.(none)')

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git config user.email "siddeshwarprasadkr@gmail.com"

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git commit -m "Initial Commit"
[master (root-commit) 7c6921c] Initial Commit
 1 file changed, 1 insertion(+)
 create mode 100644 index.txt
```

Task A4: Modify File and Commit Again

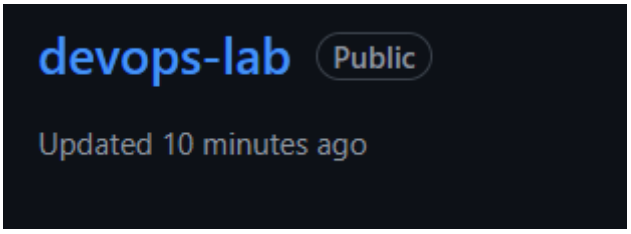
```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ echo "Learning Git is fun" >> index.txt

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git add .
warning: in the working copy of 'index.txt', LF will be replaced by CRLF the next time Git touches it

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git commit -m "Updated File"
[master ca2d648] Updated File
 1 file changed, 1 insertion(+)
```

Task A5: View History

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git log
commit ca2d648a71e501505668eff511c6cbe86a7a024c (HEAD -> master)
Author: unknown <siddeshwarprasadkr@gmail.com>
Date:   Tue Mar 31 20:45:40 2026 +0530

    Updated File

commit 7c6921c127f3bee5dddcb2be6fcb5e82d6cd6c23
Author: unknown <siddeshwarprasadkr@gmail.com>
Date:   Tue Mar 31 20:43:51 2026 +0530

    Initial Commit
```

Part B – GitHub Integration

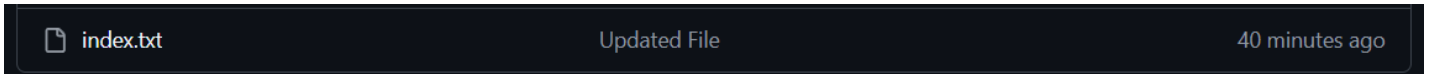
Task B1: Create GitHub Repository

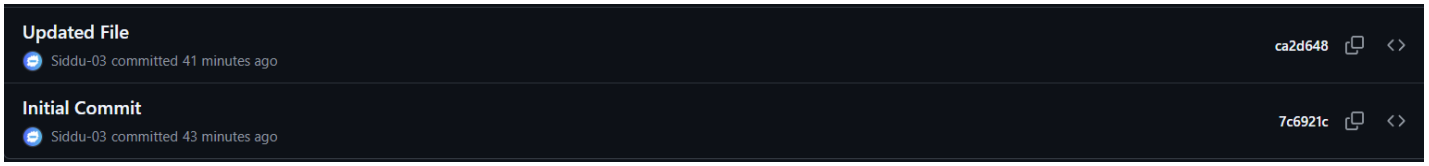


Task B2: Connect Local Repo to GitHub

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git remote add origin https://github.com/Siddu-03/devops-lab

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (master)
$ git branch -M main

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (6/6), 494 bytes | 494.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Siddu-03/devops-lab
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Task B3: Verify on GitHub





Task B4: Create a Branch

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (main)
$ git checkout -b feature-update
Switched to a new branch 'feature-update'
```

Task B5: Make Changes and Push

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (feature-update)
$ echo "New feature added" >> index.txt

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (feature-update)
$ git add .
warning: in the working copy of 'index.txt', LF will be replaced by CRLF the next time Git touches it

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (feature-update)
$ git commit -m "Feature update"
[feature-update 057f2b2] Feature update
 1 file changed, 1 insertion(+)

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (feature-update)
$ git push origin feature-update
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 306 bytes | 306.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-update' on GitHub by visiting:
remote:      https://github.com/Siddu-03/devops-lab/pull/new/feature-update
remote:
To https://github.com/Siddu-03/devops-lab
 * [new branch]      feature-update -> feature-update
```

Part C – GitHub Actions (CI Pipeline)

Task C1: Create Workflow File

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (feature-update)
$ mkdir -p .github/workflows
```

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (main)
$ touch .github/workflows/main.yml
```

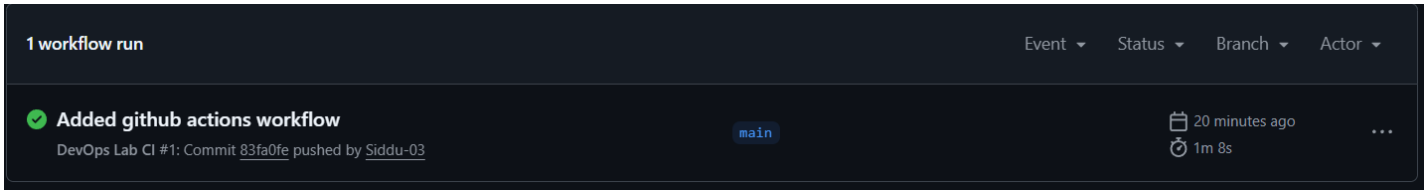
Task C2: Add Simple CI Workflow

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (main)
$ nano .github/workflows/main.yml
```

```
  GNU nano 8.7                    .github/workflows/main.yml
name: DevOps Lab CI

on:
  push:
    branches: [ "main" ]

jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v3

      - name: Display message
        run: echo "CI Pipeline Running Successfully!"
```

Task C3: Commit and Push

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (main)
$ git add .
warning: in the working copy of '.github/workflows/main.yml', LF will be replaced by CRLF the next time
 Git touches it

SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (main)
$ git commit -m "Added github actions workflow"
[main 83fa0fe] Added github actions workflow
 1 file changed, 16 insertions(+)
 create mode 100644 .github/workflows/main.yml
```

```
SIDDESHWARAPRASAD KR@LAPTOP-D2TTFURO MINGW64 ~/desktop/devops-lab (main)
$ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (5/5), 561 bytes | 561.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Siddu-03/devops-lab
   ca2d648..83fa0fe  main -> main
```

Task C4: Verify CI Execution